

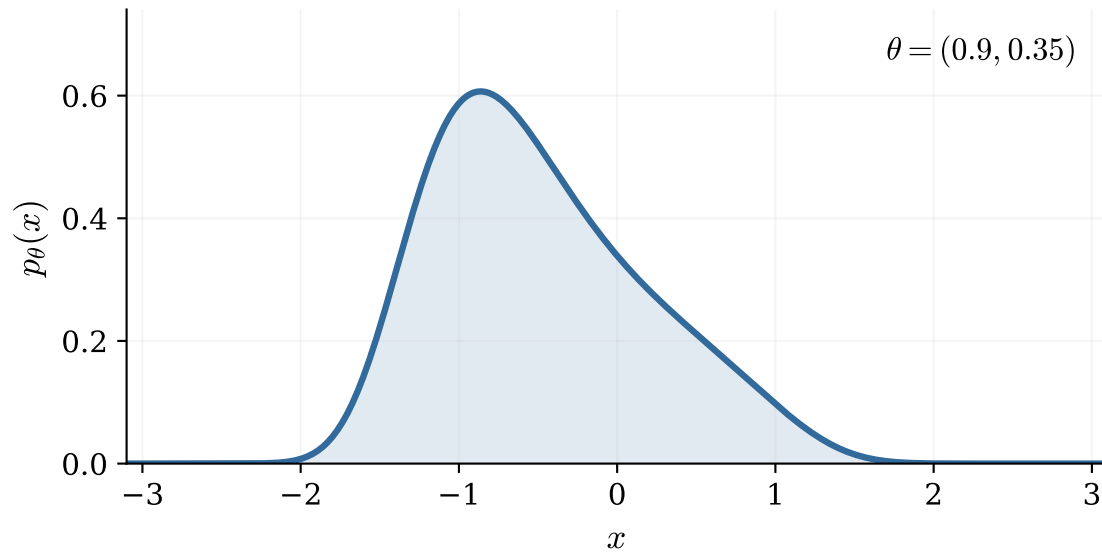
Four Gibbs families: linear and nonlinear potentials

$$p_{\theta}(x) = Z_{\theta}^{-1} e^{-U_{\theta}(x)}, \quad \theta = (\theta_1, \theta_2)$$

Linearity refers to the parameters, not to the spatial variable.

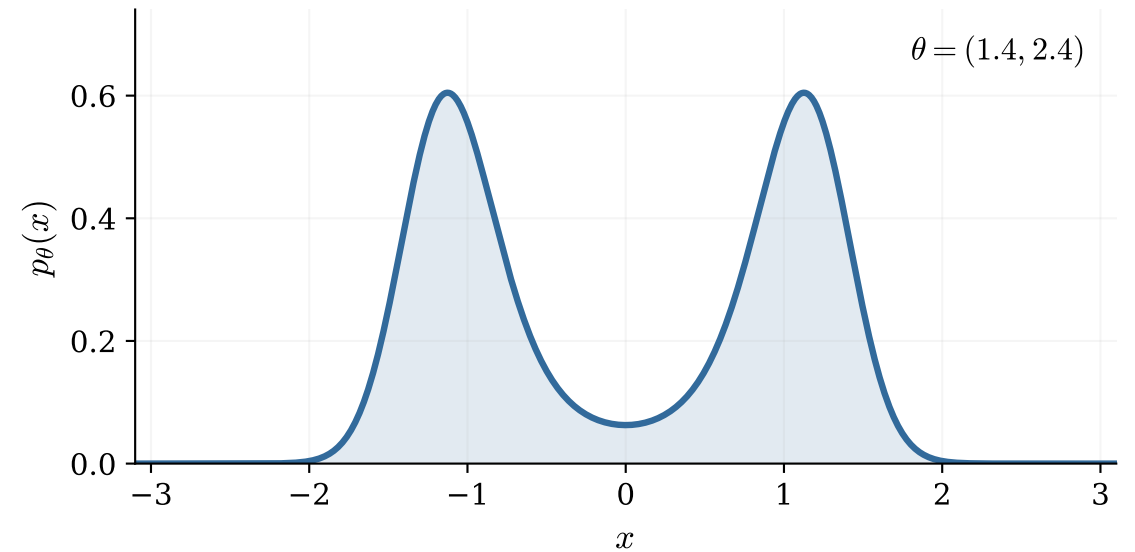
(a) Linear · 1D

$$U_{\theta}(x) = \theta_1 x + \theta_2 x^4$$



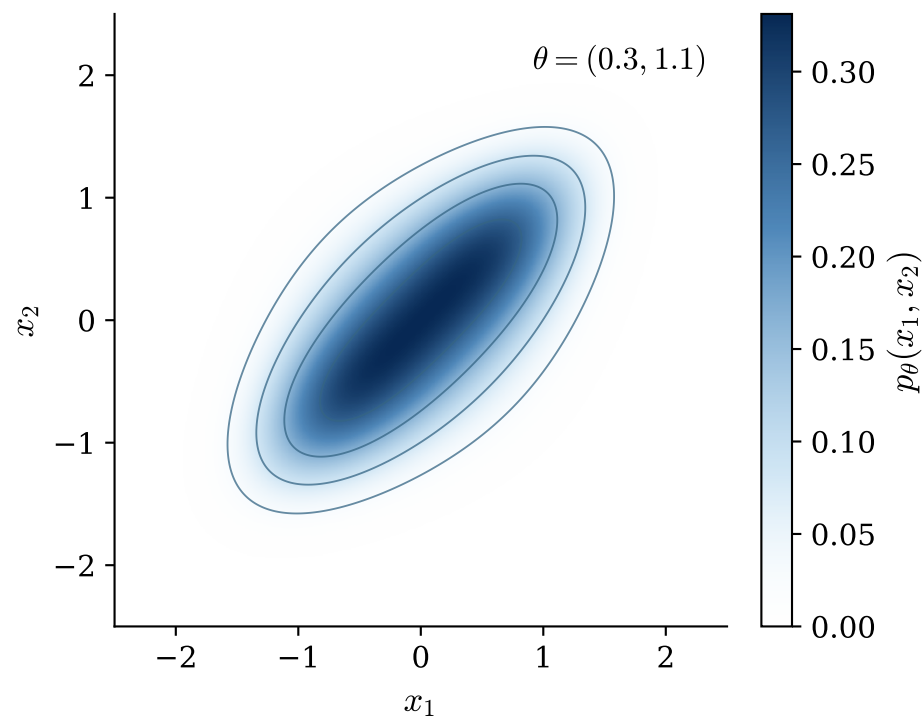
(b) Nonlinear · 1D

$$U_{\theta}(x) = \frac{x^4}{4} + \theta_1 \cos(\theta_2 x)$$



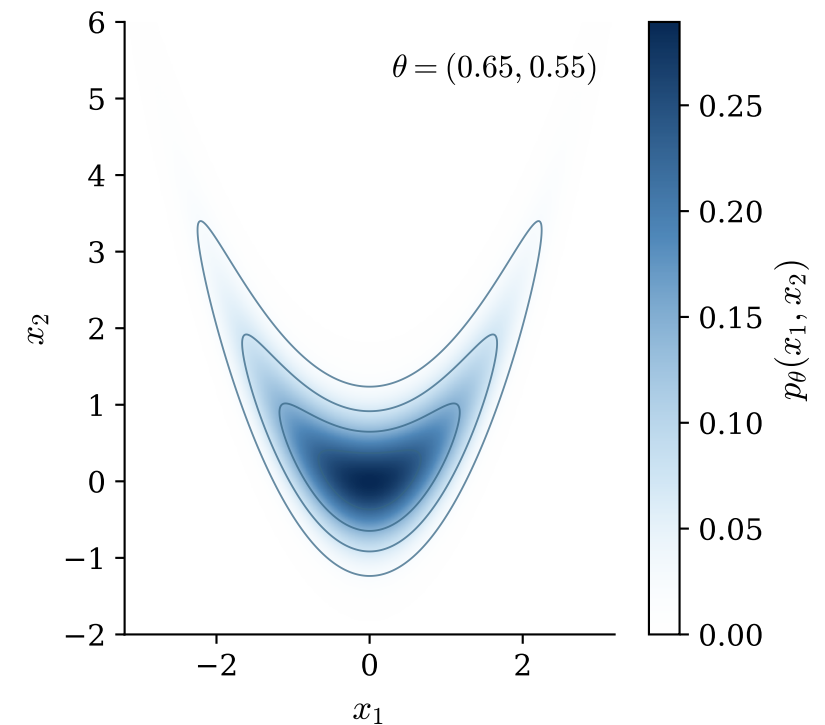
(c) Linear · 2D

$$U_{\theta}(x_1, x_2) = \theta_1(x_1^4 + x_2^4) + \theta_2(x_1 - x_2)^2$$



(d) Nonlinear · 2D

$$U_{\theta}(x_1, x_2) = \frac{x_1^2}{2} + \frac{(x_2 - \theta_1 x_1^2)^2}{2\theta_2^2}$$



All four families are defined on the whole space and have two parameters. The 2D colorbars show probability density.

1D: density curves. 2D: blue density maps with constant-density contours; the two maps have separate color scales.